Shengzhou Qiang

Buffalo, NY · sqiang@buffalo.edu · sqiang.us · github.com/Alephant6 · linkedin.com/in/sqiang

Ph.D. candidate in distributed systems — transactions, consistency protocols, and storage internals.

Available for Summer 2027 software-engineering internships

Education

University at Buffalo, SUNY — Ph.D. in Computer Science (advisor: Prof. Haonan Lu) Aug 2023 – expected Jun 2028
Advanced to candidacy 2026. Research: distributed transactions, consistency protocols, evaluation of distributed datastores.

Nanjing University of Information Science & Technology — B.S., Information and Computing Science 2019 – 2023
GPA 3.8/4.0 (top 5%); First-class Outstanding Graduate of Jiangsu Province.

Research Experience

Graduate Research Assistant, University at Buffalo — distributed transactions & databases Aug 2023 – present

SONIC — read-only transactions without replication-critical designs *first author, under review*
- Proved "replication-critical" read-only transaction designs unnecessary (SONIC & Remote-Read theorems).
- Demonstrated it in two systems — **Starry** on UW's TAPIR codebase (C++) and COPS-SNOW extended with chain replication — validated with self-built Jepsen falsifier suites; contributed fixes upstream to Jepsen.

TxJBOF — concurrency control and write-path offload for DPU + JBOF storage *lead student, in progress*
- Reproduced **Scalio** (OSDI'25, a DPU-offloaded KV store over NVMe SSDs) on CloudLab with RDMA NICs, SPDK and NVMe-oF, root-causing two artifact defects: a fixed-seed workload generator inflating reads and an -O0 build understating the write ceiling 3.4–3.6×.
- Designed a **client-side write-path offload** (one-sided RDMA index reads, RDMA-CAS publish with optimistic concurrency + group commit) that takes the DPU CPU off the write path: **2.8× upstream write throughput at 7 SSDs, near-linear in SSD count** where upstream plateaus at 4 (emulated NVMe).

Juicer — transparent scheduling layer for low tail latency *co-author, in submission*
- **Cut P99 latency by up to 75% and aborts by up to 85%** for strictly serializable databases — with zero database-code changes — by reordering in-flight transaction messages by timestamp in a gRPC-Go interceptor; **beat NCC (SOSP'23) by 54% on tail latency**.
- Implemented concurrency-control protocols (2PL wound-wait/wait-die, MVTO) in a Go distributed transactional KV testbed and built its workload generator (Zipfian, dynamic skew, read-modify-write).

EaaS — Evaluation-as-a-Service for distributed storage systems
- **Refuted EPaxos's published 3× throughput claim** with cross-region AWS measurements (1–1.5× measured) and **uncovered an EPaxos livelock at 9 replicas**, motivating a proven f·T upper bound on replica-utilization gains.
- Top contributor (**≈half of 1,800+ commits**) to the first end-to-end evaluation framework for distributed storage (C++, gRPC); built its YCSB/TPC-C workload engine and CockroachDB/Cassandra adapters.

Experience

Open-source contributor, Nervos Fiber — Rust payment-channel network on CKB 2026
- Reproduced a channel deadlock (#1561) with a self-built Jepsen suite; proposed a root-cause fix that removed it from 6,000 randomized fault schedules (1,030 ⟶ 0). Merged #1675; reported a new reconnect bug with a fix (#1689).

Teaching Assistant, UB CSE — Distributed Systems, Operating Systems, Data-Intensive Computing 2023 – 2025

Selected Publications & Manuscripts

- **SONIC** — **S. Qiang** et al., first author; under review at a top-tier systems conference, 2026.
- **Juicer** and **PRC** — co-author, in submission, 2026. **EaaS** — OSDI submission, 2025.
- 7 peer-reviewed signal-processing papers (2021–22), incl. first-author in *EURASIP J. Adv. Signal Process.*, 2021.

Skills

Languages C++, Go, Rust, Java, Python, Clojure, Shell
Distributed systems transactions & concurrency control (2PL/OCC/MVTO); replication & consensus (Paxos/Raft/EPaxos, chain replication, TAPIR); distributed KV & storage internals (CockroachDB); gRPC
Evaluation & infrastructure YCSB/TPC-C, Jepsen, P; RDMA verbs, SPDK, NVMe-oF; CloudLab, AWS, Linux